Literature-Implied Partial Answer for arXiv:1309.4684

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Status

literature_implied_answer (partial subcase). This note records a later literature theorem which answers one named family from Bendova's final open-ended question. It is not claimed as a new result of this run.

Source Question

The source is Hana Bendova, *Quantitative Grothendieck property*, arXiv:1309.4684. In Section 4, after proving that $\ell_\infty(\Gamma)$, sigma-finite $L^\infty(\mu)$, and quotients of $\ell_\infty(\Gamma)$ are 1-Grothendieck, Bendova asks whether the other Grothendieck spaces mentioned in the introduction have the quantitative version as well. The introduction names, among others, $C(K)$ for compact F-spaces, weak L^p spaces, H^∞, and Haydon's Grothendieck space not containing ℓ_∞.

Supporting Literature

The supporting paper is Jindrich Lechner, *1-Grothendieck $C(K)$ spaces*, arXiv:1511.02202. Its main theorem states:

if S is a Haydon space, then $C(S)$ is 1-Grothendieck.

Lechner defines a Haydon space as a totally disconnected compact space whose algebra of clopen subsets has the subsequential completeness property. The paper also points out that Haydon had constructed such a space K with $C(K)$ not containing an isomorphic copy of ℓ_∞. Therefore the specific Haydon example named in Bendova's introduction has an affirmative quantitative answer, with constant 1.

Lechner further records that $C(K)$ is 1-Grothendieck whenever K is sigma-Stonean, and in particular whenever K is extremally disconnected. These are additional $C(K)$-side positive cases, but they do not by themselves cover all compact F-spaces.

Scope and Remaining Cases

The identification gives the following partial answer to arXiv:1309.4684:

$$\boxed{\text{Haydon-space } C(K) \text{ examples are 1-Grothendieck.}}$$

In particular, Haydon's Grothendieck space without ℓ_∞ satisfies Bendova's quantitative Grothendieck property with the optimal constant allowed by the definition.

The packet does not settle the other families listed by Bendova: general compact F-space $C(K)$ spaces, weak L^p spaces, or H^∞. Lechner explicitly leaves the other examples from his introduction open after the Haydon-space theorem. During this bounded pass, local registry searches and web searches for close phrases around 1-Grothendieck, quantitative Grothendieck, H^∞, weak L^p, compact F-spaces, and Haydon spaces found Lechner's theorem as the decisive supporting result and did not locate a complete later answer for the remaining Bendova examples.

References

1. H. Bendova, *Quantitative Grothendieck property*, arXiv:1309.4684.

2. J. Lechner, *1-Grothendieck $C(K)$ spaces*, arXiv:1511.02202.